

Term Project: Climate Change and Carbon Pricing

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ECON 204: Contemporary Economic Issues

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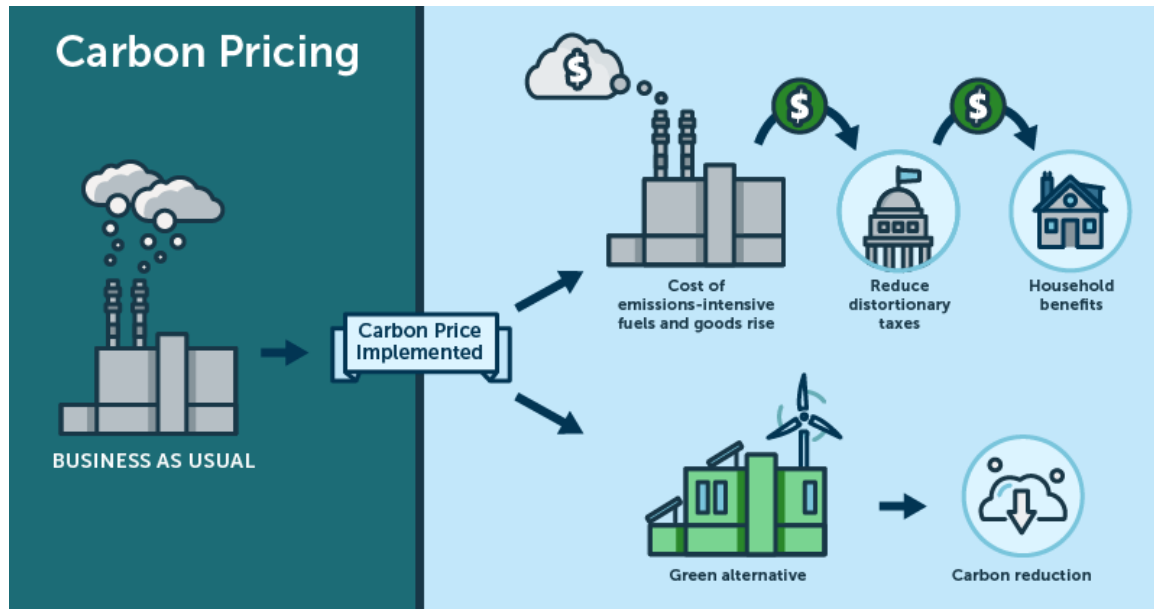
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Climate change is an ongoing issue globally, and governments worldwide are trying to find ways to lessen the impact of climate change by mitigating greenhouse gas emissions. The earth is warming due to emissions and greenhouse gases, and the effect of this warming is expected to be severe. In Canada and other places, carbon pricing aims to reduce greenhouse gas emissions; with this, Canada hopes that businesses and households move towards using cleaner energy sources. In this paper, we will discuss the implementation of carbon pricing in Canada and its economic impact on businesses and households.

Carbon pricing is a combination of federal and provincial policies that puts a cost on creating pollution to encourage businesses and people to reduce their greenhouse gas emissions. In Canada, this includes some provinces' federal carbon tax and a cap-and-trade system. This section examines key economic ideas, research and data about how well carbon pricing works and how it affects people and businesses.

Carbon pricing is based on negative externalities, meaning pollution harms others without the polluter paying for it (Pigou, 1920). To fix this, the government charges polluters a Pigouvian tax, making them pay for the damage they cause. Another important idea is the Coase Theorem (Coase, 1960), which says that if people can bargain quickly, they can solve pollution problems independently. However, governments usually have to step in in real life because bargaining is often expensive and complicated.

Exhibit A:



Several studies have looked at how carbon pricing works in Canada. Rivers and Schaufele (2015) studied British Columbia's carbon tax, which was introduced in 2008. They found that gasoline use dropped by 12% in four years, and the economy grew. Another study by the Canadian Institute for Climate Choices (2021) found that carbon pricing has helped lower emissions in some provinces. However, extra policies like investing in clean energy are also needed.

Carbon pricing did raise costs for businesses, especially in industries that use much energy or rely on gas. The Canadian Chamber of Commerce (2022) said that industries like manufacturing, farming and transportation worry about competing with businesses in countries that have lower carbon costs, making them more cost-effective as these businesses often have to compete with businesses in the states and other countries. Some businesses pass the extra costs on to customers, while others invest in cleaner technology to cut pollution. A study by Clean Prosperity (2023) found that money from carbon pricing is being used to help businesses switch

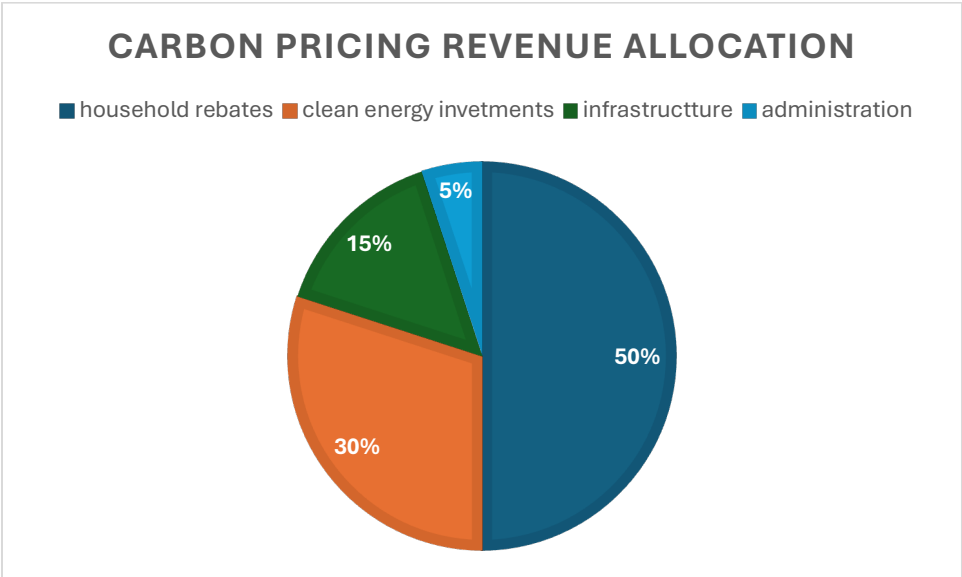
to greener solutions. This is a good solution, but it requires a lot of capital and time, and it is not easy to reshape a business around cleaner energy.

One big concern about carbon pricing in every Canadian's mind is that it makes things more expensive for people. An independent study by the Parliamentary Budget Officer (2022) found that the federal carbon tax increases fuel and heating costs. However, at the same time, most people get rebates that cover the extra costs, so this should not be a big problem.

Approximately 90% of the revenue is returned to individuals through the Canada Carbon Rebate and the other 10% goes back to farmers, small-and medium-enterprises and Indigenous governments. The report says that lower- and middle-class households get more money back than they spend, which makes the policy fair instead of unfair. However, rural households may pay more because they rely more on gas and have fewer public transportation options and services, unlike big cities. (Canada Energy Regulator, 2023).

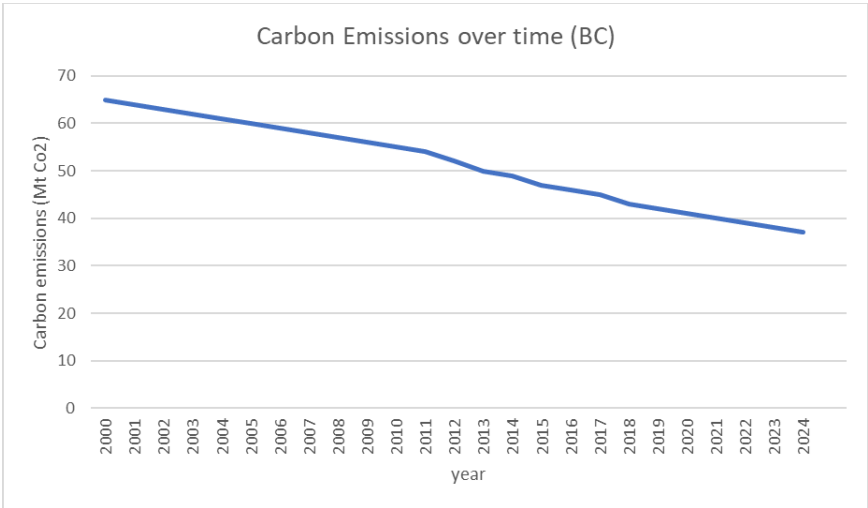
Research shows that carbon pricing has helped lower emissions in Canada, especially in places like British Columbia. However, the economic impact is different for different groups. Most households get enough rebates to cover the costs, but businesses in high-pollution industries still face challenges. More research is needed to understand the long-term effects of how well other government policies support carbon pricing and whether this strategy is even a good fit for Canadian society. We could learn from other countries who face similar problems but manage to solve them using different strategies.

Exhibit B:



Carbon Pricing Revenue Allocation in Canada

Exhibit C:

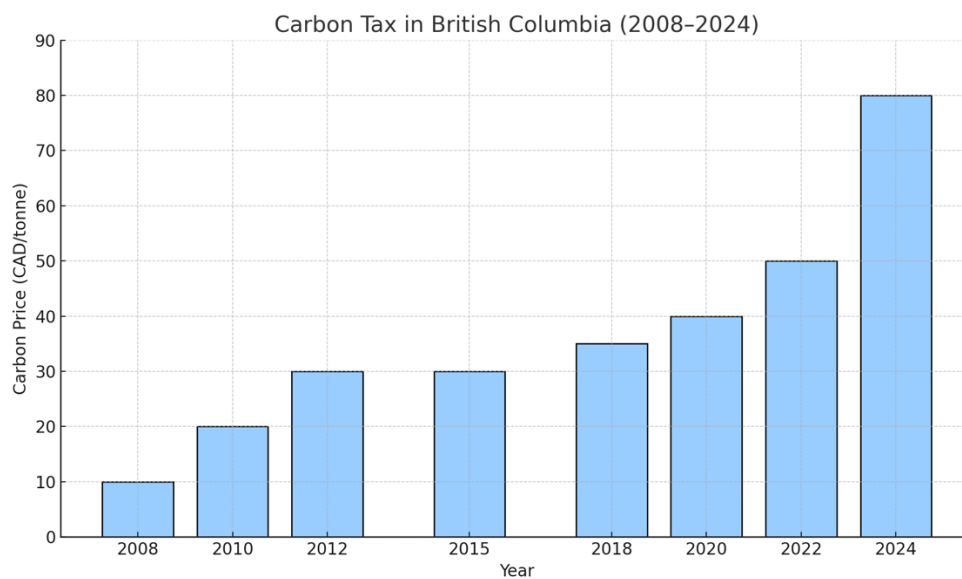


Carbon Emissions in British Columbia (2000-2025)

In Canada, all provinces have different carbon pricing policies. In 2008, British Columbia introduced a carbon tax, starting at CAD 10 per tonne and a scheduled annual increase of CAD 5 per tonne until the tax reached CAD 30 per tonne (Harrison, 2013). In 2008, the government mentioned that money collected from the carbon tax would be returned through reductions in

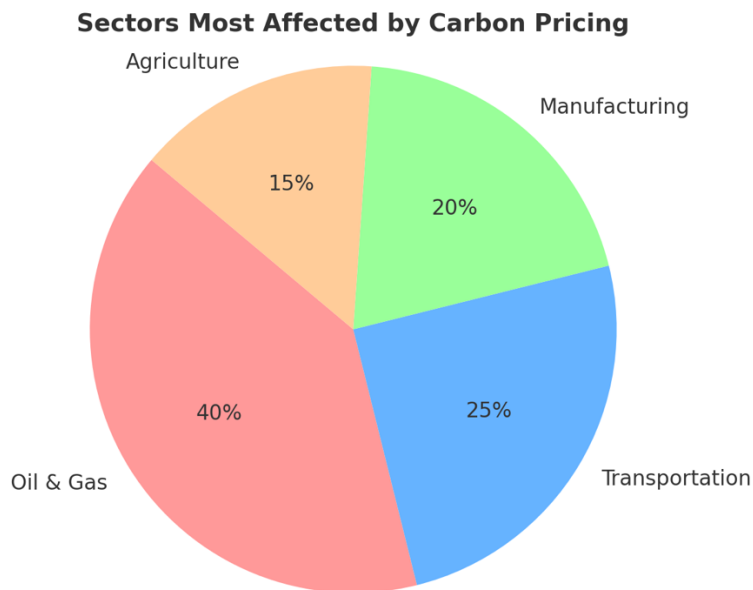
personal and corporate taxes. This raised a concern for many Northern rural communities. The carbon tax was a start for Canada to reduce emissions. As of April 2024, the carbon tax rate is CAD 80 per tonne of emissions (Government of British Columbia, 2024). Since the carbon tax initiative in British Columbia was introduced, it has been successful; in BC, fossil fuel consumption has declined rapidly compared to any other Canadian Province (Jahanandish, 2020). British Columbia had implemented the carbon tax in hopes of reducing greenhouse gas emissions and lessening the effect of climate change. The hope was that businesses and households would lessen the use of fossil fuels and emissions to save money and help towards climate change. The carbon tax in British Columbia was initially controversial, especially in Northern rural communities, but now has proven to be successful in reducing per capita fuel consumption while maintaining strong economic growth. In 2019, the federal government of Canada implemented a nationwide carbon tax to ensure all provinces and territories in Canada were doing their part to reduce fossil fuels and emissions.

Exhibit D:



The nationwide carbon tax was implemented to reduce fossil fuels and emissions, but it came with increased costs and competition for businesses in Canada. Carbon tax incentivizes businesses to move to cleaner energy and reduce carbon footprint. Carbon taxes encourage more efficient use of resources by motivating businesses and individuals to adopt energy-efficient practices and technologies. The carbon tax in Canada had a significant impact on some industries and sectors. There were short-term and long-term impacts of the carbon tax. With the nationwide carbon tax, sectors like oil, electricity, natural gas liquids, and related products would encounter a price increase. Carbon tax affects industries like oil and gas more than others because of the high carbon footprint. Of 236 commodities in the Canadian economy, 73 would experience a price increase of approximately 1% or more (McKittrick et al., 2019). Sectors such as renewable energy and green technology are experiencing growth and job creation due to increased demand for cleaner energy. Carbon tax increases production costs for some countries, making Canadian products more expensive than others in the global market, leading to a decreased demand for Canadian products in the global market. Since prices will increase for Canadian products, this will lead to a decrease in demand, but with markets with low elasticities, the changes may be small (McKittrick et al., 2019). Elastic markets (luxury goods) will see a more significant decline than inelastic markets (essentials) and a slight decline in demand despite the price increases. With an increase in carbon tax, there is a threat of carbon leakage, which is businesses moving their operations to places with a lower carbon tax rate. In energy-intensive sectors, labour demand will decline, decreasing wages and salaries (McKittrick et al., 2019). This will result from increased carbon tax, resulting in companies struggling to compete globally, lower profit margins, and increased costs. However, In the short run, carbon tax would increase costs for businesses, but in the long run, they will notice the savings and move towards cleaner industries.

Exhibit E:



The carbon tax will have a positive and negative impact on households in Canada. This impact can depend on various factors, such as Canada's urban vs rural living areas and low-income versus high-income households. In 2024, gasoline prices increased by 17.6 cents due to carbon pricing, which is responsible for 0.15 percent of a given year's inflation increase, according to the Bank of Canada (Clean Energy Canada, 2024). Carbon pricing has caused price increases for many goods and services, affecting households. Provinces with high emissions intensity have higher carbon costs, provinces like Alberta and Saskatchewan have high emission intensity, which leads to higher carbon costs because there is higher transportation fuel use and natural gas use in households in Alberta and Saskatchewan (Winter, Dolter, & Fellows, 2023). Low-income households have been impacted by higher prices, leading them to improvise on some things. High-income households usually use more gasoline, natural gas, and electricity than low-income households. With increased carbon tax, high-income households will shift from buying luxury goods and non-essential items, and low-income households will focus on buying

essential items. There will be an increased cost of living because gasoline prices increase with carbon tax, household heating costs increase, and services such as flights will increase.

The reaction to increased cost of living will depend on many factors, including household income and urban vs rural communities. With the increased cost of living, demand for luxury goods and non-essential items will fall, and households will focus on essentials. Many households may be eligible for rebates, returning carbon tax revenue to households. Rebates will depend on family size, income, and location and are also given quarterly. Households with low–middle income typically receive more rebates than high-income households. Urban households will have less impact than rural households because of community facilities. With the carbon tax, Canada aims to encourage households to reduce fossil fuel use and encourage the purchase of e-vehicles through rebates as incentives. Households that adopt energy-efficient habits will save money in the long run and save money through rebates. The goal of Canada is to reduce emissions while still protecting low-income households from financial burdens.

According to research and evidence, the carbon tax has been successful in Canada because of the decline in fossil fuels and emissions. The government can incentivize citizens to reduce their carbon footprint through policies that will balance the interests of the public while still aiming to reduce the carbon footprint. The government has many rebates and incentives for energy-efficient technology, but it should also enhance carbon tax rebates for low-income households to help with raised prices and costs. The government should add more rebates for households trying to be more energy efficient by adding energy-efficient technology. With rebates, households will be more inclined to implement energy-efficient technology into their households and will not have to worry as much about financial burden; the government will also have to ensure they are providing extra support for rural communities as they have higher energy costs.

Along with rebates, the government should enhance subsidies and incentives for businesses and households to transition to renewable energy sources, including solar, wind, and hydroelectric power. Another policy is needed to avoid carbon leakage, which will help prevent businesses from moving operations to countries with lower carbon taxes. The government must create trade policies restricting businesses from committing carbon leakage. The government can provide rebates or tax relief for highly competitive businesses with countries without carbon taxes. Overall, the government will need to monitor and re-adjust policies according to the country's changing dynamics. They must regularly review carbon tax rates and rebates to ensure they align with the country's inflation and economic conditions.

In conclusion, carbon pricing in Canada has played a big part in reducing greenhouse gas emissions while affecting businesses and households in different ways. Research shows carbon tax policies like those in British Columbia have successfully lowered fossil fuel consumption without hindering economic growth. The economic impact varies depending on the industry; some industries have faced increased production costs and competition. Households have experienced higher fuel and heating costs, but rebates have helped with the financial burden for most Canadians. Moving forward, Canada should create a balance between reducing emissions and maintaining economic stability. Regular policy reviews, adaptive measures, and learning from global best practices will be essential in optimizing carbon pricing strategies. With this approach, Canada can continue to lead in climate action while balancing economic resilience and stability.

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